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Enhancing Zero-Shot Relation Extraction through Staged Interaction with Large Language Models
通过与大语言模型进行分阶段交互，提升零样本关系抽取性能

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***Abstract*—Zero-shot Relation Triplet Extraction (ZeroRTE) is a challenging yet valuable task that extracts relation triplets from unstructured texts for new relation types, significantly reducing the time and effort needed for data labeling. With the advancement of the zero-shot capabilities of large language models, the performance of many zero-shot tasks has been further improved only simply by interacting with large language models (LLMs). In this work, we transform the zero-shot triplet extraction task into a two-stage chat with LLMs. Specifically, in the first stage, we prompt the LLMs to perform Named Entity Recognition (NER). In the second stage, we prompt the LLMs to perform Relation Classification (RC) using the results from the first stage. Experiments on Wiki-ZSL and FewRel datasets show the efficacy of Relation Prompt for the ZeroRTE task. Notably, our method significantly outperforms strong baselines, achieving an impressive 15.89% increase in F1 scores, particularly on Wiki-ZSL with 15 unseen relations.**

***摘要*—零样本关系三元组抽取（ZeroRTE）是一项具有挑战性但极具价值的任务，它从非结构化文本中提取适用于新关系类型的关系三元组，从而显著减少数据标注所需的时间和精力。随着大语言模型零样本能力的进步，许多零样本任务的性能仅通过与大语言模型（LLM）交互即可进一步提升。在本研究中，我们将零样本三元组抽取任务转化为与大语言模型的两阶段对话流程。具体而言，在第一阶段，我们引导大语言模型执行命名实体识别（NER）任务；在第二阶段，我们利用第一阶段的结果，引导大语言模型执行关系分类（RC）任务。在 Wiki- ZSL 和 FewRel 数据集上的实验表明，关系提示法在 ZeroRTE 任务中具有显著效果。值得注意的是，我们的方法显著优于强基线模型，F1 分数提升了 15.89%，尤其在包含 15 个未见关系的 WikiZSL 数据集上表现突出。**

***Index Terms*—Relation Triplet Extraction, LLMs, Zero-Shot, Prompt, Instruction-tuning.**

***索引术语*—关系三元组提取、大型语言模型、零样本学习、提示、指令微调。**

I. Introduction
I. 引言

Relation Triplet Extraction(RTE) aims to extract a full triplet, a complete triplet, namely (head_entity, relation_type, tail_entity), from unstructured text. This process has applications in areas such as knowledge graph construction and question answering [1]. Recently, GraphRAG [2] has captured the spotlight for its innovative approach and potential. GraphRAG leverages structural information across entities to enable more precise and comprehensive retrieval and facilitating more accurate responses. The construction of the knowledge graph forms the foundation of GraphRAG, where the quality of the relation triplet extraction directly impacts its performance [3]. However, existing RTE approaches often require large datasets of annotated samples, which are costly to create, and are limited by a fixed set of relations [4]. Hence, many researchers have recognized the value of information extraction (IE) techniques with zero/few-shot methods [5] [6]

关系三元组抽取（RTE）旨在从非结构化文本中提取完整的三元组，即（头实体、关系类型、尾实体）。该过程在知识图谱构建与问答等领域具有重要应用价值[1]。近年来，GraphRAG[2]凭借其创新方法与潜在应用前景备受关注。GraphRAG利用实体间的结构化信息，实现更精准、更全面的检索，进而生成更准确的响应。知识图谱的构建是GraphRAG的基础，其中关系三元组抽取的质量直接决定其性能[3]。然而，现有RTE方法通常需要大量标注样本数据集——其构建成本高昂，且受限于固定关系集合[4]。因此，众多研究者已认识到零/少样本学习方法在信息抽取（IE）领域的重要价值[5] [6]。

[7]. Under the zero-shot setting, the relation sets used during training and testing are disjoint, as shown in the Figure 1. ZeroRTE models are trained on samples with a limited number of seen relation types and are expected to generalize to extract triplets with new or unseen relation types.

[7]在零样本设置下，训练阶段与测试阶段所使用的关系集互不相交，如图 1 所示。ZeroRTE 模型在仅包含有限数量已知关系类型的样本上进行训练，并期望能够泛化，以提取包含新或未见关系类型的三元组。

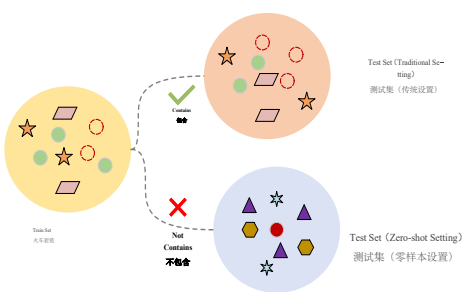


Fig. 1. Zero-shot relation triplet extraction.
图 1. 零样本关系三元组提取。

With the widespread application of large language models, their zero-shot capabilities have garnered increasing attention [8]. Using carefully designed prompts has become a popular method to unlock the potential of these models for zero-shot tasks [9] [10] [11] [12]. Chain of Thought (CoT) prompting [13], which involves guiding the LLMs through a step-by-step breakdown of complex problems, has proven effective in addressing intricate issues. Although it is non-trivial to tackle ZeroRTE by decomposing it into two sub-tasks, the challenges of **How to plan the two-stage tasks?** and **How to design the prompt to enhance the ZeroRTE ability of LLMs?** still need to be addressed. RSED [14] and ChatIE [15] have decomposed ZeroRTE into Relation Selection(RS) and Entity boundary Detection, which may increase the propagation of errors, as illustrated in the Figure 2. This is because Relation Selection, the first stage, requires a deeper understanding of semantics compared to NER.

随着大语言模型的广泛应用，其零样本能力已引起越来越多的关注[8]。采用精心设计的提示语已成为释放这些模型在零样本任务中潜力的常用方法[9] [10] [11] [12]。思维链（CoT）提示法[13]——即引导大语言模型逐步分解复杂问题——已被证明能有效应对复杂问题。尽管将ZeroRTE分解为两个子任务并非易事，但**如何规划这两个阶段任务？**以及**如何设计提示语以增强大语言模型的ZeroRTE能力？**这两个挑战仍需解决。 RSED[14]与ChatIE[15]已将ZeroRTE分解为关系选择（RS）与实体边界检测，但如图2所示，这种分解可能会加剧误差传播，因为第一阶段——关系选择——相较于命名实体识别（NER）需要更深层的语义理解。

In our work, ZeroRTE is divided into two straightforward subtasks: NER and RC. By decomposing the complex ZeroRTE task into two more manageable subtasks, as opposed to Relation Selection and Entity Boundary Detection, error propagation is alleviated, which makes it easier for the model to solve the RTE. For each stage, we consider designing the prompts from three aspects: 1)Task Description, 2)Background knowledge, 3)Output Format.

在本研究中，ZeroRTE 被划分为两个简单的子任务：NER 与 RC。通过将复杂的 ZeroRTE 任务分解为两个更易处理的子任务——而非采用“关系选择”与“实体边界检测”的组合方式——可有效缓解误差传播，从而助力模型更便捷地解决 RTE 任务。针对每个子任务阶段，我们分别从以下三个维度设计提示词：1) 任务描述；2) 背景知识；3) 输出格式。

Following the method described above, extensive experiments were conducted on the ZeroRTE tasks. Specifically, our approach advances the state-of-the-art RSED [14] model on two ZeroRTE datasets, achieving an improvement of up to 15.89% in the F1 score compared to the previous best model on Wiki-ZSL.

按照上述方法，我们在 ZeroRTE 任务上进行了大量实验。具体而言，我们的方法在两个 ZeroRTE 数据集上优化了最先进的 RSED[14]模型，与之前在 Wiki- ZSL 上的

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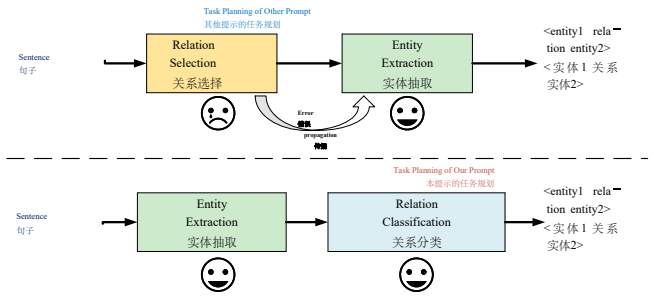


Fig. 2. Task Planning of different prompt

图 2. 不同提示词的任务规划

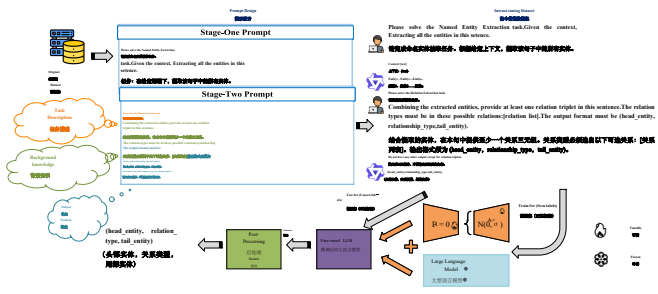


Fig. 3. Overall Architecture of Models.

图 3. 模型整体架构。

最佳模型相比，F1 分数最高提升了 15.89%。

The main contributions of this paper are summarized as follows:

本文的主要贡献总结如下：

(1) We introduced an innovative two-stage framework that leverages large language models (LLMs) for zero-shot relation triplet extraction. Our approach effectively combines NER and RC tasks, enhanced by the carefully designed prompt, which significantly reduces error propagation.

(1) 我们提出了一种创新的两阶段框架，利用大语言模型（LLMs）实现零样本关系三元组抽取。该方法有效融合了命名实体识别（NER）与关系抽取（RC）任务，并辅以精心设计的提示词，从而显著降低了误差传播。

(2) We compared the impact of different two-stage task planning strategies and various prompt designs on the ZeroRTE task, providing a detailed analysis of their effects.

(2) 我们比较了不同两阶段任务规划策略及多种提示设计对 ZeroRTE 任务的影响，并对其效应进行了详尽分析。

(3) Experiments on two datasets demonstrate that our proposed method is state-of-the-art (SOTA) in the field of ZeroRTE.

(3) 在两个数据集上进行的实验表明，我们提出的方法在 ZeroRTE 领域达到了当前最佳性能（SOTA）。

II. Method

II. 方法

A. Task Formulation

A. 任务制定

Let $D = \{S, < head_entity, R, tail_entity >\}$denotes the whole dataset, consisting of the input sentences S, the output triplets $< head_entity, R, tail_entity >$ where R is the set of relation labels. $D = D_s \cup D_u$, where D_s, D_u refer to the seen and unseen datasets respectively. The model is trained on D_s and evaluated on D_u. $R = R_s \cup R_u$ is predefined, comprising the seen relation label set $R_s = \{r_1^s, \cdots, r_n^s\}$ and unseen relation label set $R_u = \{r_1^u, \cdots, r_m^u\}$, where $n = |R_s|$ and $m = |R_u|$ are the number of seen and unseen relation labels respectively. $R_s \cap R_u = \emptyset$, R_s and R_u are disjoint. One sentence $s \in S$ contains one or more triplets.

令 $D=\{S, <head_entity, R, tail_entity>\}$ 表示完整数据集，包含输入句子 S 以及输出三元组 $<head_entity, R, tail_entity>$，其中$R$为关系标签集合。$D=D_s \cup D_u$，其中 D_s, D_u分别表示已见数据集与未见数据集。模型在D_s上训练，并在D_u上评估。$R=R_s \cup R_u$为预定义集合，包含已见关系标签集$R_s=\{rs1, \cdots, rsn\}$与未见关系标签集 $R_u=\{ru1, \cdots, rum\}$，其中$n=|R_s|$且$m=|R_u|$ 分别表示已见与未见关系标签的数量。$R_s \cap R_u=\emptyset$ ，且R_s与R_u互不相交。一个句子$s \in S$包含一个或多个三元组。

B. Overview

B. 概述

To tackle the ZeroRTE task, we propose a two-stage dialogue framework based on the LLMs. The framework divides the task into two phases: NER and RC, with carefully designed prompts for each stage. Using these prompts, we created an instruct-tuning dataset. We then fine-tuned the LLM using the Low-Rank Adaptation (LoRA) method [16] based on this instruct-tuning dataset.

为应对 ZeroRTE 任务，我们提出了一种基于大语言模型（LLM）的两阶段对话框架。该框架将任务划分为两个阶段：命名实体识别（NER）与关系抽取（RC），并为每个阶段设计了精心构建的提示词。利用这些提示词，我们构建了一个指令微调数据集；随后，基于该数据集，我们采用低秩适配（LoRA）方法[16]对大语言模型进行了微调。

C. Two-stage Prompts Design

C. 两阶段提示词设计

We have designed a two-stage prompts for zero-shot tasks, aimed at decomposing the complex relation triplet extraction problem into two more manageable subtasks. This approach is intended to improve the performance of large models in relation extraction tasks. The overall architecture of the prompt design is detailed in Figure 3.

针对零样本任务，我们设计了一套两阶段提示策略，旨在将复杂的关系三元组抽取问题分解为两个更易处理的子任务。该方法旨在提升大模型在执行关系抽取任务时的性能。提示设计的整体架构详见图 3。

1) Stage-One: Named Entity Recognition: ChatIE [15] and RSED [14] transform Relation Triplet Extraction into relation category selection and relation generation tasks. Our experiments reveal that large language models (LLMs) are prone to significant biases during the relation type selection phase when using this approach, which adversely affects the final performance of relation extraction. This issue arises because selecting relation types requires a nuanced understanding of semantics to make accurate choices. In contrast, entity recognition is a relatively straightforward task that involves identifying relevant entities in the text.

1) 第一阶段：命名实体识别： ChatIE[15]与 RSED[14]将关系三元组提取任务转化为关系类别选择与关系生成任务。我们的实验表明，当采用此方法时，大型语言模型（LLMs）在关系类型选择阶段容易产生显著偏差，进而对关系提取的最终性能产生不利影响。这一问题源于选择关系类型需要对其语义进行细致理解，才能做出准确选择。相比之下，实体识别则是一项相对简单的任务，主要涉及从文本中识别出相关实体。

Consequently, in our research, we have replaced the relation type selection task with entity recognition and designed the prompts as illustrated in Figure 3.

因此，在本研究中，我们将关系类型选择任务替换为实体识别任务，并设计了如图 3 所示的提示语。

2) Stage-Two: Relation Classification and Triplets Generation: This stage is the core phase of relation triplet extraction, responsible for combining the results of the previous entity extraction round, performing relation classification, and generating triplets that conform to the specified format. The design of prompt in this stage is mainly considered from three aspects:

2) 第二阶段：关系分类与三元组生成： 本阶段是关系三元组提取的核心环节，负责整合前一轮实体提取的结果，执行关系分类，并生成符合指定格式的三元组。本阶段提示词的设计主要从以下三个方面考虑：

(1) Task Description: In this stage, the prompt is designed to clearly specify the task of relation triplet extraction. The prompt should guide the model to generate triples that adhere to the specified format.

(1) 任务描述：在本阶段中，提示词旨在明确定义关系三元组提取任务，并引导模型生成符合指定格式的三元组。

(2) Background Information: Instructions should be provided on how to combine the entities extracted in the previous phase with the provided context. The prompt should direct the model to perform relationship classification based on this information and to generate relation triplets in the specified format. Each triplet should consist of a subject entity, a relationship label, and an object entity.

(2) 背景信息：应提供关于如何将前一阶段提取的实体与给定上下文相结合的指导说明。提示词应引导模型基于该信息执行关系分类任务，并以指定格式生成关系三元组。每个三元组均由主体实体、关系标签及客体实体组成。

(3) Output Format: The prompt must clearly define the expected output format for the generated relation triplets. It should specify the structure and organization of each triplet, including the order of entities and the representation of the relationship label. This ensures that the model produces output that conforms to the desired format.

(3) 输出格式：提示词必须清晰定义所生成关系三元组预期的输出格式，并明确每个三元组的结构与组织方式，包括实体间的排列顺序以及关系标签的表示形式。此举可确保模型生成的输出符合预期格式要求。

Following the three design principles mentioned above, we have designed the prompt for the second stage as illustrated in Figure 3.

基于上述三项设计原则，我们已针对第二阶段设计了提示词，如图 3 所示。

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D. LLMs Fine-tuning
D. 大语言模型微调

After finalizing the prompt design, we integrated it with the dataset to create the instruct-tuning dataset for our work. Subsequently, we fine-tuned the LLMs using LoRA based on this instruction-tuning dataset.

在完成提示词设计后，我们将该设计与数据集相结合，构建了用于本研究的指令微调数据集。随后，我们基于该指令微调数据集，采用 LoRA 对大语言模型进行微调。

Our model involves two training tasks: named entity recognition and relation triplets generation. We train the model in a multi-task manner. For named entity recognition, the training objective is to minimize the cross entropy loss:

本模型包含两个训练任务：命名实体识别与关系三元组生成。我们采用多任务方式训练该模型。对于命名实体识别任务，训练目标是最小化交叉熵损失函数：

$$\mathcal{L}_{ent} = -\sum_{c=1}^{C} y_{o,c} \log(p_{o,c}) \tag{1}$$

For relation triplets generation, we also adopt the cross entropy loss, and the entity loss is defined as:

在生成关系三元组时，我们同样采用交叉熵损失函数，而实体损失函数则定义为：

$$\mathcal{L}_{rel} = -\sum_{t=1}^{T} y_{o,t} \log(p_{o,t}) \tag{2}$$

Here, $\mathcal{L}_{ent}$ denotes the loss function of the NER task. $\mathcal{L}_{rel}$ denotes the loss function of the relation triplets generation task. C represents the number of tokens for the entity to be predicted. T represents the number of tokens for the triplets to be predicted. In the NER phase, $y_{o,c}$ and $p_{o,c}$ represent the ground truth values and the predictions of the LLMs, respectively. In the relation triplets generation phase, $y_{o,t}$ and $p_{o,t}$ represent the ground truth values and the predictions of the LLMs, respectively.

此处，$\mathcal{L}_{ent}$表示NER任务的损失函数；$\mathcal{L}_{rel}$表示关系三元组生成任务的损失函数；C代表待预测实体的标记数；T代表待预测三元组的标记数。在NER阶段，$y_{o,c}$与$p_{o,c}$分别表示真实标注值及LLM的预测结果；在关系三元组生成阶段，$y_{o,t}$与$p_{o,t}$分别表示真实标注值及LLM的预测结果。

We treat each loss equally and the model learns to minimize $\mathcal{L} = \mathcal{L}_{rel} + \mathcal{L}_{ent}$ jointly.

我们对每个损失项一视同仁，模型则共同学习以最小化L=$\mathcal{L}_{rel} + \mathcal{L}_{ent}$。

TABLE I
表 I
RESULTS COMPARED WITH BASELINE MODELS
(**BOLD** VALUES INDICATE THE BEST MODEL, WHILE UNDERLINE REPRESENTS THE SECOND-BEST.)
结果与基准模型对比（粗体数值表示最佳模型，下划线表示次优模型）

Unseen Labels 未显示标签	Method 方法	Wiki-ZSL			Fewrel		
		P(%)	R(%)	F1(%)	P(%)	R(%)	F1(%)
m=15	RelationPrompt [17]	26.19	32.12	28.85	17.73	23.20	20.08
	PAED [18]	-	-	-	20.68	23.39	21.95
	ZETT [19]	-	-	21.21(±0.35)	-	-	24.39(±0.37)
	RSED [14]	25.37	33.80	28.98	27.00	23.55	25.16
	Ours(Llama)	56.47	36.45	44.30	**31.28**	24.87	**27.71**
	我们的（Llama）						
	Ours(Qwen)	**57.31**	**36.87**	**44.87**	30.28	**25.03**	27.41
	我们的（Qwen）						
m=10	RelationPrompt [17]	30.20	32.31	31.19	21.59	28.68	24.61
	PAED [18]	-	-	-	23.31	27.42	25.15
	ZETT [19]	-	-	24.87(±0.32)	-	-	31.28(±0.54)
	RSED [14]	27.09	**39.09**	32.00	30.89	29.90	30.39
	Ours(Llama)	56.79	36.76	44.63	34.98	**32.45**	33.67
	我们的（Llama）						
	Ours(Qwen)	**60.77**	37.76	**46.57**	**35.46**	32.17	**33.74**
	我们的（Qwen）						
m=5	RelationPrompt [17]	29.11	31.00	30.01	20.80	24.32	22.34
	PAED [18]	-	-	-	25.79	34.54	29.47
	ZETT [19]	-	-	31.17 (±0.69)	-	-	33.71 (±0.42)
	RSED [14]	38.14	**36.84**	**37.48**	43.91	34.97	38.93
	Ours(Llama)	42.47	20.27	27.40	50.77	**52.86**	51.79
	我们的（Llama）						
	Ours(Qwen)	**42.52**	23.27	30.09	**53.10**	52.66	**52.88**
	我们的（Qwen）						

III. EXPERIMENTS
III. 实验

A. Experimental Settings
A. 实验设置

For the task of zero-shot relation triplets extraction, we fine-tune the Qwen1.5-14B-Chat [20] proposed by Alibaba Cloud and Llama2-13B-Chat [21] developed by Meta. We use the following two datasets for our experiments. FewRel [22] is hand-annotated for few-shot relation extraction, but we made it suitable for the zero-shot setting after data splitting into disjoint relation label sets for training and testing. Wiki-ZSL [23] is constructed through distant supervision over Wikipedia articles and the Wikidata knowledge base. For each dataset, we varied the number of unseen labels, setting it to $m \in \{5, 10, 15\}$, while the remaining labels were treated as seen during training in our experiments.

对于零样本关系三元组提取任务，我们分别微调了阿里云提出的 Qwen1.5-14B-Chat[20]和 Meta 开发的 Llama2-13B-Chat[21]。我们在实验中使用了以下两个数据集：FewRel[22]是专为少样本关系提取而人工标注的数据集，但我们在将其划分为训练集和测试集两个互不相交的关系标签集后，使其适用于零样本设置；Wiki- ZSL[23]则是通过对维基百科文章和 Wikidata 知识库进行远程监督构建而成。对于每个数据集，我们在实验中调整未见过标签的数量，设定为$m \in \{5,10,15\}$，而其余标签在训练过程中均被视为已见过标签。

To evaluate multiple triplet extraction, we use the Micro F1 metric which is standard in structured prediction tasks, and report the precision (P.) and recall (R.). The experimental results represent the mean values obtained from five independent replicates.

为评估多三元组提取性能，我们采用结构化预测任务中的标准指标——Micro F1 指标，并报告精确率（P.）与召回率（R.）。实验结果均取自五次独立重复实验的平均值。

B. Experimental Results
B. 实验结果

To demonstrate the effectiveness of our proposed method in the ZeroRTE task, we compare it with several existing ZeroRTE models.

为验证本文提出的方法在 ZeroRTE 任务中的有效性，我们将该方法与若干现有 ZeroRTE 模型进行了对比。

• **RelationPrompt** [17] fine-tuned the BART [24] on the synthetic relation sentences generated by prompting language models.

• **RelationPrompt**[17]在通过提示语言模型生成的合成关系句上对 BART[24]进行了微调。

• **PAED** [18] presented a generation-based framework for zero-shot PAED. A novel HNS strategy and a Meta-VAE sampler with CSC are presented to enhance the performance of this model.

• **PAED**[18]提出了一种基于生成的零样本 PAED 框架。本文还提出了一种新型 HNS 策略以及一种结合 CSC 的 Meta- VAE 采样器，以提升该模型的性能。

• **ZETT** [19] resolved the Zero-shot Triplet extraction by Template infilling based on an end-to-end generative language model.

• **ZETT**[19]基于端到端的生成式语言模型，通过模板填充解决了零样本三元组提取问题。

• **RSED** [14] proposed a method with potential candidate relation selection and entity boundary detection directly utilizing the semantics of unseen relations to tackle the ZeroRTE task.

• **RSED**[14]提出了一种方法，该方法通过直接利用未见关系的语义，进行潜在候选关系选择与实体边界检测，以解决 ZeroRTE 任务。

The experimental results are reported in Table I. Our framework consistently outperforms existing methods. For instance, with 15 unseen labels, our framework improves the F1 score by 2.25% on the Fewrel dataset and by 15.89% on the Wiki-ZSL dataset, compared to the RSED, which is the previous state-of-the-art method. In conclusion, our proposed method demonstrates a superior ability to leverage the potential of LLMs in the RTE task.

实验结果见表I。我们的框架始终优于现有方法。例如，在拥有15个未见标签的情况下，与先前最先进的方法 RSED 相比，我们的框架在Fewrel数据集上使F1分数提升了2.25%，在WikiZSL数据集上则提升了15.89%。综上所述，我们提出的方法在利用大型语言模型解决RTE任务方面展现出卓越的能力。

However, it is important to note that when the number of unseen labels is 5, our method's performance on the Wiki-ZSL dataset is not as strong as that of the RSED method. This discrepancy may be attributable to the distribution characteristics of the Wiki-ZSL dataset. Additionally, our observations suggest that with fewer labels, there is a higher likelihood of generating triplets that fall outside the predefined relations, which could contribute to the observed performance decline. Despite this, our method still outperforms other models. Addressing this issue will be a key focus of our future work.

然而，需要注意的是，当未见的标签数量为5时，我们的方法在WikiZSL数据集上的性能不如 RSED 方法。这一差异可能归因于Wiki- ZSL 数据集的分布特性。此外，我们的观察表明，当标签数量较少时，生成不符合预定义关系的三元组的可能性更高，这可能导致性能下降。尽管如此，我们的方法仍优于其他模型。解决这一问题将成为我们未来工作的重点方向。

IV. ANALYSIS
四、分析

In this section, we focus on two main research questions proposed in the Introduction Section. The experiments in this section are all conducted using the Qwen model and verified on the Fewrel dataset.

在本节中，我们聚焦于引言部分提出的两个核心研究问题。本节中的所有实验均使用 Qwen 模型开展，并均在 Fewrel 数据集上进行了验证。

A. How to plan the two-stage tasks?
A. 如何规划两阶段任务?

To compare the impact of different two-stage task settings on model performance, we designed other two-stage models with different content for the experiments. Taking the FewRel as the test dataset, we conducted experiments under a different

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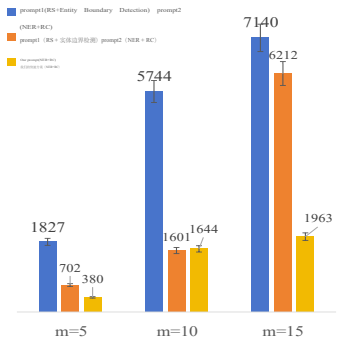


Fig. 4. Number of Predictions Outside Pre-defined List.

number of unseen label settings, and the experimental results are shown in Table II.

为比较不同两阶段任务设置对模型性能的影响，我们在实验中设计了采用不同内容的其他两阶段模型。以 FewRel 作为测试数据集，我们在不同数量的未见标签设置下进行了实验，实验结果如表 II 所示。

- **prompt1(RS+Entity boundary Detection)**
- **prompt1（RS+实体边界检测）**

This prompt still employs a two-stage task framework. However, unlike the prompt used in this paper, we have modified the first stage to perform RS instead of the NER task.

本提示仍采用两阶段任务框架。但与本文所使用的提示不同，我们已对第一阶段进行改进，使其执行 RS 任务而非 NER 任务。

TABLE II
表 II
RESULTS WITH DIFFERENT PROMPT
使用不同提示词的结果

Unseen Labels 未显示标签	Method 方法	P(%)	R(%)	F1(%)
m=15	*w.t. prompt1* *相对于 prompt 1*	**37.20**	11.91	18.04
	w.t. prompt2 *相对于 prompt 2*	17.73	23.20	20.08
	Our prompt 我们的提示	30.28	**25.03**	**27.41**
m=10	*w.t. prompt1* *相对于 prompt 1*	**51.59**	9.26	15.70
	w.t. prompt2 *相对于 prompt 2*	35.58	28.24	31.49
	Our prompt 我们的提示	35.46	**32.17**	**33.74**
m=5	*w.t. prompt1* *相对于 prompt 1*	**67.72**	32.37	43.80
	w.t. prompt2 *相对于 prompt 2*	50.85	41.09	45.45
	Our prompt 我们的提示	53.10	**52.66**	**52.88**

The results indicate that using **prompt1**, which involves selecting relation types first and then generating triplets, performs poorly. As illustrated in Figure 4, the number of predictions outside the predefined relation list is higher with **prompt1**. This observation underscores the critical importance of task selection in a two-stage approach for relation triplet extraction.

实验结果表明，采用**prompt1**（即先选择关系类型，再生成三元组）的方法性能较差。如图 4 所示，使用**prompt 1**时，超出预定义关系列表的预测数量更多。这一观察结果凸显了在关系三元组提取的双阶段方法中，任务选择至关重要。

The poor performance observed with the Prompt1 task setup can be attributed to its initial phase of relation selection, which demands a deep semantic understanding. This increased complexity heightens the risk of errors being propagated, leading to a higher incidence of incorrect relationship types or instances that fall outside the predefined list.

在 Prompt1 任务设置中观察到的性能不佳，可归因于其初始阶段的关系选择环节——该环节需要深度的语义理解能力。这种复杂性的增加加剧了错误传播的风险，从而导致错误关系类型或超出预定义列表的实例出现频率升高。

Consequently, we conclude that the two-stage task setup involving NER followed by RC is more effective in minimizing predictions outside the predefined relation list, thereby improving performance on the Zero-RTE task.

因此，我们得出结论：采用先进行命名实体识别（NER）、随后进行关系抽取（RC）的两阶段任务架构，更能有效减少对预定义关系列表外预测结果的生成，从而提升 Zero-RTE 任务的性能。

B. How to design the prompt to enhance the ZeroRTE ability of LLMs?
B. 如何设计提示词以提升 LLM 的 ZeroRTE 能力?

- **prompt2(NER+RC)** This prompt adopts the same two-stage task setup as this paper, but its task description is cumbersome.
- **prompt2(NER+RC)**该提示采用了与本论文相同的两阶段任务设置，但其任务描述较为冗长。

From Figure 4 and Table II, we observed that **prompt2** shows slightly inferior performance compared to the prompt used in this paper. Subsequently, we conducted ablation experiments on each component of our designed prompts. As shown in Table III, the removal of any component leads to a decline in performance.

根据图 4 和表 II，我们观察到**prompt2**的性能相较于本文采用的提示语略有下降。随后，我们对所设计提示语的每个组成部分进行了消融实验。如表 III 所示，移除任一组成部分均会导致性能下降。

Therefore, we can conclude that a more concise and clear prompt is more conducive to the LLM's understanding and completion of downstream tasks.

因此，我们可以得出结论：更简洁、更清晰的提示语更有利于大语言模型理解并完成下游任务。

TABLE III
表 III
DIFFERENT DESIGN OF THE PROMPT
提示语的不同设计

Unseen Labels 未显示标签	Method 方法	P(%)	R(%)	F1(%)
m=15	*w.o.Background Information* *无。背景信息*	**37.21**	21.46	27.22
	w.o.Output Format *无输出格式*	25.78	5.59	9.18
	Ours 我们的	30.28	**25.03**	**27.41**
m=10	*w.o.Background Information* *无。背景信息*	**42.18**	25.65	31.90
	w.o.Output Format *无输出格式*	33.52	10.37	15.84
	Ours 我们的	35.46	**32.17**	**33.74**
m=5	*w.o.Background Information* *无。背景信息*	52.80	28.91	37.36
	w.o.Output Format *无输出格式*	51.70	16.12	24.58
	Ours 我们的	**53.10**	**52.66**	**52.88**

C. Other Ablation Study
C. 其他消融研究

TABLE IV
表 IV
ABLATION STUDY RESULTS
消融研究结果

Unseen Labels 未显示标签	Method 方法	P(%)	R(%)	F1(%)
m=15	*w.o.first-stage prompt* *无第一阶段提示*	29.42	24.69	26.84
	w.o.LoRA	22.06	8.51	12.29
	Ours 我们的	**30.28** **30.28**	**25.03**	**27.41**
m=10	*w.o.first-stage prompt* *无第一阶段提示*	33.30	26.93	29.78
	w.o.LoRA	17.18	4.76	7.45
	Ours 我们的	**35.46** **35.46**	**32.17**	**33.74**
m=5	*w.o.first-stage prompt* *无第一阶段提示*	51.29	40.49	45.25
	w.o.LoRA	43.59	11.09	17.68
	Ours 我们的	**53.10** **53.10**	**52.66**	**52.88**

We first removed the prompt from the first stage, using only the second stage's prompt and input text. Subsequently, we eliminated the fine-tuning module and relied solely on the designed prompts for model inference.

我们首先从第一阶段中移除提示词，仅使用第二阶段的提示词与输入文本；随后，我们移除了微调模块，在模型推理过程中仅依赖预先设计的提示词。

The results show a substantial decline in model performance, indicating that both the two-stage prompts approach and LoRA fine-tuning are crucial for enhancing the performance of LLMs in the ZeroRTE task.

实验结果表明，模型性能显著下降，这说明两阶段提示方法及 LoRA 微调对于提升 LLM 在 ZeroRTE 任务中的性能至关重要。

V. CONCLUSIONS
V. 结论

This paper addresses the Zero-shot Relation Triplet Extraction (ZeroRTE) task and introduces an innovative two-stage conversational approach to improve the extraction of relation triplets from unstructured texts involving previously unseen relation types. The experimental results demonstrate that our method significantly outperforms established baselines.

本文探讨了零样本关系三元组抽取（ZeroRTE）任务，并提出一种创新的两阶段对话式方法，旨在提升从包含未见关系类型的非结构化文本中抽取关系三元组的性能。实验结果表明，本方法相较于现有基准模型具有显著优势。

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